

Name: _____

This practice exam covers Chapters 9 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. James is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 88.9-cm. After filling, the pressure inside the tank is 145 atm and the temperature is 310. K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

2. The reaction $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \longrightarrow \text{CaCl}_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ is carried out in a sealed, initially evacuated vessel with a volume of 9.9 L. After 5.95 g of HCl completely reacts while all other reactants are present in excess at 18. °C, determine the pressure of CO_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

3. The reaction $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 11 mL of oxygen gas reacts completely, what volume of nitric oxide is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

4. A fixed amount of gas is initially at 6.02 °C and 4.35 atm. It is later found to have a volume that is 29% smaller than its initial volume and a pressure of 7.55 atm. Determine the final temperature. Report your answer in SI units and justify your answer.

ANSWER: _____

5. A sample of an unknown gas is transferred into a dry gas bulb. The sample has a mass of 3.672 g, occupies 0.750 L at 150.0 kPa and 298.15 K. Possible identities are: SO_2 , CO_2 , Cl_2 , Ne, CH_4 , NO, UF_6 , He, H_2S , H_2 , Ar, F_2 , Xe, Kr, CO, HBr, C_2H_2 , H_2O , SF_6 . Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

6. A gas sample is trapped inside a syringe with its tip sealed. Its initial pressure is 2.0×10^2 kPa and its initial volume is 9.4 L. The piston is moved until the gas occupies 1.0 L while the temperature remains constant. Determine the final pressure. Report your answer in SI units and justify your answer.

ANSWER: _____

7. A gas pushes against a rectangular piston measuring 12-cm by 7.8-cm. The gas pressure is 7.65 kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

8. A 13.6-L tank at 112.°C and 1.36 atm contains 0.38 mol of chlorine trifluoride (ClF_3) and 0.383 mol of carbon monoxide (CO). Determine the mole fraction of chlorine trifluoride. Justify your answer.

ANSWER: _____

9. A sample of 20. g of ammonium hydrogen carbonate (NH_4HCO_3) reacts completely with excess HCl in a sealed container. Before the reaction begins, the container holds 0.20 mol of argon (Ar). No gas escapes, and the inert gas does not react. Determine the mole fraction of carbon dioxide after the reaction is complete. Justify your answer.

ANSWER: _____

10. A liquid is boiling at 59°C in a chamber where the external pressure is 83 kPa. Determine whether the liquid's vapor pressure at this temperature is less than, equal to, or greater than the external pressure. Report the external pressure in SI units and justify your answer.

ANSWER: _____

11. An evacuated flask is filled with CCl_4 having a measured volume of 6.39 L. After the gas reaches 7.4°C, the pressure inside is measured to be 0.400 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

12. A rigid cylinder containing an ideal gas is heated while no gas enters or leaves the cylinder. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

13. Consider the reaction $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \longrightarrow 2\text{HBr}(\text{g})$. If 4.66cm^3 of H_2 gas is completely consumed at the same temperature and pressure, what volume of HBr gas is produced? Justify your answer.

ANSWER: _____

14. A 9.0-L tank at 73.4°C and 0.97 atm contains 0.45 mol of hydrogen bromide (HBr) and 0.21 mol of nitrogen trifluoride (NF_3). Determine the mole fraction of nitrogen trifluoride. Justify your answer.

ANSWER: _____

15. A 5.76 g sample of carbon dioxide gas (CO_2) is contained in a sealed reaction vessel at 11.6°C and 1.1 atm. Determine the sample volume assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

16. A pressurized gas cylinder is connected to a large evacuated balloon. The cylinder has an internal diameter of 5.0-cm and an internal height of 19.2-cm. After the gas is released until atmospheric pressure is reached, the total volume occupied by the gas is 2.80 L. Determine the initial pressure inside the cylinder. Report your answer in SI units and justify your answer.

ANSWER: _____

17. A sealed flask is evacuated and then filled with an unknown gas. The sample has a mass of 6.821 g, occupies 1.250 L at 101.3 kPa and 293.15 K. Possible identities are: Xe, CH_4 , F_2 , Ar, H_2O , SO_2 , Cl_2 , SF_6 , NO, Ne, H_2S , HBr , CO_2 , Kr, CO, UF_6 , C_2H_2 , H_2 , He. Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

18. A gas mixture has a total pressure of 0.964 atm. The mole fraction of He is 0.350. Determine the partial pressure of the gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

19. A vertical cylinder containing gas has a piston of diameter 15-cm. The cylinder is 9.25-cm tall. A constant downward force of 47-N acts on the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

20. The reaction $\text{N}_2\text{O}_4(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 5.6 cm^3 of dinitrogen tetroxide is consumed, what volume of nitrogen dioxide is produced or consumed? Justify your answer.

ANSWER: _____

21. A research balloon contains 2.14 L of nitrogen at 27°C while indoors. It is then taken outside, where the air temperature is -21°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Report your answer in SI units and justify your answer.

ANSWER: _____

22. A rigid 11-L tank at $124.^\circ\text{C}$ contains only methane (CH_4) and dinitrogen difluoride (N_2F_2). Assume both gases behave ideally. The total pressure is 1.03 atm. The tank contains 6.96 g of methane and 8.82 g of dinitrogen difluoride. Determine the pressure that would remain if only the heavier gas stayed in the tank. Justify your answer.

ANSWER: _____

23. A gas cylinder is opened into a large empty flexible reservoir. The cylinder has an internal diameter of 6.5-cm, an internal height of 26.1-cm, and initially contains gas at 3.86 atm. The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Report your answer in SI units and justify your answer.

ANSWER: _____

24. A 527-mL sample of R-22 refrigerant (CHClF_2) is initially at 392 K. Its boiling point is $-41.^\circ\text{C}$ (232.15 K). The pressure is then increased to 2.0 times its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest $^\circ\text{C}$. Justify your answer.

ANSWER: _____

25. A sample of CS_2 is transferred into a volumetric gas flask having a measured volume of 12.0 L. After the gas reaches 3.7°C , the pressure inside is measured to be 0.76 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

26. Cl_2 is introduced into an evacuated flask having a measured volume of 5.8 L. After the gas reaches $19.^\circ\text{C}$, the pressure inside is measured to be 0.59 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

27. A gas is reported to be C_2H_6 . Its density is measured as $1.6 \times 10^3 \frac{\text{g}}{\text{m}^3}$ at 130 kPa and $24.^\circ\text{C}$. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

28. The reaction $\text{CS}_2(\text{g}) + 3 \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2 \text{SO}_2(\text{g})$ occurs at constant temperature and pressure. If 6.5 cm^3 of CO_2 gas reacts completely, what volume of CS_2 gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

29. O_2 is collected over water. The total pressure of the gas sample is 761 torr, and the vapor pressure of water is 29 torr. Determine the partial pressure of the dry gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

30. Mark is preparing a deep-sea submersible. A rigid cylindrical air tank has an internal diameter of 28.0-cm and an internal height of 126-cm. After filling, the pressure inside the tank is 158 atm and the temperature is 283 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

31. A gas-filled syringe has an internal diameter of 13-cm. The plunger is pushed inward with a constant force of 37.8-N. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

32. SiF_4 is introduced into an evacuated collection bulb having a measured volume of 12 L. After the gas reaches $12.5^\circ C$, the pressure inside is measured to be 0.328 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

33. A sample of an unknown gas is transferred into a dry gas bulb. The sample has a mass of 1.130 g, occupies 1.000 L at 100.0 kPa and 298.15 K. Possible identities are: He, NO, UF_6 , SF_6 , HBr, SO_2 , H_2 , CH_4 , CO_2 , F_2 , Kr, H_2S , C_2H_2 , H_2O , CO, Xe, Cl_2 , Ar, Ne. Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

34. An unknown compound is completely vaporized at $110^\circ C$. At a pressure of 1.02 atm, the gas has a density of 1.84 g/L. Determine the molar mass of the compound. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

35. The vapor pressure of an unknown compound is measured at 57°C . Additional liquid of the same substance is added while the temperature is held constant and liquid remains present before and after the addition. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

36. A vertical cylinder containing gas has a piston of diameter 10.3-cm. The cylinder is 7.17-cm tall. A constant downward force of 34-N acts on the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

37. A flexible latex balloon contains 2.8 L of nitrogen at 25.9°C while indoors. It is then taken outdoors, where the air temperature is -15.0°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Report your answer in SI units and justify your answer.

ANSWER: _____

38. A fixed amount of gas changes from 20.4°C to 54°C while its pressure changes from 6.2 atm to 18 atm. Determine the final volume as a percentage of the initial volume. Justify your answer.

ANSWER: _____

39. The reaction $\text{MgCO}_3(\text{s}) \longrightarrow \text{MgO}(\text{s}) + \text{CO}_2(\text{g})$ is carried out in a sealed, initially evacuated vessel with a volume of 19.5 L. After 5.9 g of MgCO_3 completely reacts while all other reactants are present in excess at 5.2°C , determine the pressure of CO_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

40. Consider the reaction $2\text{N}_2\text{H}_4(\text{g}) + \text{N}_2\text{O}_4(\text{g}) \longrightarrow 3\text{N}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g})$. If 5.24 cm^3 of N_2H_4 gas is completely consumed at the same temperature and pressure, what volume of N_2 gas is produced? Justify your answer.

ANSWER: _____

41. A sample of 21.0g of iron (Fe) reacts completely with excess HCl in a sealed container. Before the reaction begins, the container holds 0.15mol of argon (Ar). No gas escapes, and the inert gas does not react. Determine the mole fraction of hydrogen after the reaction is complete. Justify your answer.

ANSWER: _____

42. A rigid 14.0-L tank at 140°C contains only acetylene (C₂H₂) and propane (C₃H₈). Assume both gases behave ideally. The total pressure is 1.36 atm, and the mole ratio of acetylene:propane is 1.45 : 1. Determine the pressure produced by the second gas. Justify your answer.

ANSWER: _____

43. The reaction $2\text{H}_2\text{S} + 3\text{O}_2 \longrightarrow 2\text{SO}_2 + 2\text{H}_2\text{O}$ occurs with all gaseous substances at the same temperature and pressure. During the reaction, 9.1 mL of O₂ reacts completely. Determine the corresponding volume of H₂O. Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

44. An unknown gas has a mass of 39.0g and occupies 7.61 L at 140kPa and 7.82°C. Determine the molar mass of the gas. Justify your answer. Report your answer in SI units.

ANSWER: _____

45. An unknown gas has a mass of 21.6g and occupies 5.94L at 164kPa and 38.°C. Determine the molar mass of the gas. Justify your answer. Report your answer in SI units.

ANSWER: _____

46. The reaction $2\text{C}_2\text{H}_6(\text{g}) + 7\text{O}_2(\text{g}) \longrightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 9.5mL of carbon dioxide gas reacts completely, what volume of water vapor is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

47. A flexible weather balloon contains 2.5L of neon at -8.1°C while indoors. It is then taken outside, where the air temperature is 18.9°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Report your answer in SI units and justify your answer.

ANSWER: _____

48. The reaction $\text{Mg(s)} + 2\text{HCl(aq)} \longrightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ is carried out in a sealed, initially evacuated vessel with a volume of 33L. After 6.3 g of Mg completely reacts while all other reactants are present in excess at 11.1°C , determine the pressure of H_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

49. A vertical cylinder has an internal diameter of 11.0-cm and contains gas beneath a movable piston. The gas column is 9.8-cm tall, and the gas pressure is measured to be 7.57kPa. Determine the force exerted by the gas on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

50. An unknown compound is completely vaporized at $137.^{\circ}\text{C}$. At a pressure of 0.913atm, the gas occupies 930mL and has a mass of 1.35 g. Determine the molar mass of the compound. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

51. Consider the reaction $2\text{HF(g)} + \text{SiF}_4\text{(g)} \longrightarrow \text{H}_2\text{SiF}_6\text{(g)}$. If $11.\text{cm}^3$ of HF gas is completely consumed at the same temperature and pressure, what volume of H_2SiF_6 gas is produced? Justify your answer.

ANSWER: _____

52. The reaction $2\text{NO}_2\text{(g)} + \text{F}_2\text{(g)} \longrightarrow 2\text{NO}_2\text{F(g)}$ occurs with all gases measured at the same temperature and pressure. If 14.9mL of fluorine gas reacts completely, what volume of nitryl fluoride is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

53. William is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 84.7-cm. After filling, the pressure inside the tank is 206 atm and the temperature is 286 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

54. The vapor pressure of a liquid is measured at 46°C. The sample is then heated to 57°C in a sealed container while liquid remains present. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

55. A gas is confined beneath a circular piston with a diameter of 10.2-cm. The gas pressure is measured to be 16.6 kPa. Determine the force exerted by the gas on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

56. A 624-mL sample of neon (Ne) is initially at 118 K. Its boiling point is -246.°C (27.15 K). The pressure is then reduced to 67% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest °C. Justify your answer.

ANSWER: _____

57. A rigid container is evacuated and then refilled to the same temperature and pressure with a different ideal gas whose molar mass is smaller. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

58. A rigid 9.3-L tank at 83.1°C contains only acetylene (C_2H_2) and sulfur tetrafluoride (SF_4). Assume both gases behave ideally. The tank contains 8.79 g of acetylene and 7.38 g of sulfur tetrafluoride. Determine the decimal fraction of the total pressure that would remain if all of the sulfur tetrafluoride were removed while the temperature and volume remained unchanged. Justify your answer.

ANSWER: _____

59. A sample of H_2 is collected over water. The partial pressure of the dry gas is 706 torr, and the vapor pressure of water is 17.1 torr. Determine the total pressure of the gas sample. Report your answer in SI units. Justify your answer.

ANSWER: _____

60. A mercury barometer reads 736.4 torr. Determine the atmospheric pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

61. A sample of 11.3 g of lithium carbonate (Li_2CO_3) reacts completely with excess HCl in a sealed container. Before the reaction begins, the container holds 0.436 mol of argon (Ar). No gas escapes, and the inert gas does not react. Determine the mole fraction of carbon dioxide after the reaction is complete. Justify your answer.

ANSWER: _____

62. Oxygen gas is collected over water. The measured pressure is 747 torr, and the vapor pressure of water is 29.9 torr. Determine the partial pressure of the oxygen gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

63. Oxygen gas is collected over water. The measured pressure is 764.2 torr, and the vapor pressure of water is 27.7 torr. Determine the partial pressure of the oxygen gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

64. A 13-L tank at $94.^\circ\text{C}$ and 1.22 atm contains 8.4 g of nitrosyl chloride (NOCl) and 11 g of ammonia (NH_3). Determine the mole fraction of nitrosyl chloride. Justify your answer.

ANSWER: _____

65. The reaction $2\text{NO}_2(\text{g}) \longrightarrow 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$ occurs with all gases measured at the same temperature and pressure. If $10. \text{cm}^3$ of nitrogen dioxide is consumed, what volume of oxygen is produced or consumed? Justify your answer.

ANSWER: _____

66. A laboratory apparatus contains gas beneath a circular movable piston of diameter 5.25-cm. The gas column is 5.55-cm high before any compression. A constant downward force of 58-N is applied to the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

67. The reaction $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 4.3 cm^3 of nitrogen dioxide is consumed, what volume of nitric oxide is produced or consumed? Justify your answer.

ANSWER: _____

68. A gas is sealed inside a frictionless piston. Initially the gas has a pressure of 256 kPa and occupies 11.9 L. The pressure is then increased to 540 kPa while the temperature remains constant. Determine the final volume. Report your answer in SI units and justify your answer.

ANSWER: _____

69. A fixed amount of gas is initially at 34.2°C . After its temperature is increased to 76°C and its volume becomes 27% smaller than its initial volume, its pressure is measured to be 11 atm. Determine the initial pressure. Report your answer in SI units and justify your answer.

ANSWER: _____

70. The reaction $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{H}_2\text{O}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If $10.\text{ cm}^3$ of hydrogen is consumed, what volume of water vapor is produced or consumed? Justify your answer.

ANSWER: _____

71. Additional ideal gas is introduced into a rigid container while the temperature is held constant. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

72. A sample of 19 g of sodium peroxide (Na_2O_2) reacts completely with water in a sealed container. Before the reaction begins, the container holds 0.590 mol of nitrogen (N_2). No gas escapes, and the inert gas does not react. Determine the mole fraction of oxygen after the reaction is complete. Justify your answer.

ANSWER: _____

73. An unknown compound is completely vaporized at 88.2°C . At a pressure of 1.3 atm, the gas occupies 1.32×10^3 mL and has a mass of 1.1 g. Determine the molar mass of the compound. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

74. A 15 g sample of propylene gas (C_3H_6) is contained in a sealed reaction vessel at 16°C and 1.02 atm. Calculate the volume occupied assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

75. A closed-end mercury manometer connected to a gas sample has a height difference of 81 mmHg. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

76. The vapor pressure of a volatile solvent is measured at 29°C . The sample is then heated to 60°C in a sealed container while liquid remains present. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

77. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 151.8 MPa. The window of a circular viewport can safely withstand a maximum force of 4.35 MN. Determine the largest allowable diameter of the viewport. Report your answer in SI units and justify your answer.

ANSWER: _____

78. A gas mixture contains CO_2 , CH_4 , and N_2 . The total pressure is 0.9105 atm. The mole fractions of CO_2 and CH_4 are 0.200 and 0.250, respectively. Determine the partial pressure of N_2 . Report your answer in SI units. Justify your answer.

ANSWER: _____

79. The reaction $2\text{HCl} + \text{Na}_2\text{CO}_3 \longrightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ occurs with all gaseous substances at the same temperature and pressure. If the reaction consumes 8.8 mL of CO_2 , determine the volume of HCl . Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

80. A 12.4 g sample of carbon dioxide gas (CO_2) is contained in a sealed reaction vessel at 17.0°C and 1.2 atm. Determine the sample volume assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

81. A gas is contained beneath a square piston with sides of 6.33-cm. The gas pressure is measured to be 8.5 kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

82. A mercury barometer reads 752 torr. Determine the atmospheric pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

83. Consider the reaction $\text{CS}_2(\text{g}) + 3\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{SO}_2(\text{g})$. If 12.7 cm^3 of CS_2 gas is completely consumed at the same temperature and pressure, what volume of SO_2 gas is produced? Justify your answer.

ANSWER: _____

84. A gas mixture contains 0.26 mol of chlorine trifluoride (ClF_3) and 0.47 mol of iodine pentafluoride (IF_5). Determine the mole fraction of chlorine trifluoride. Justify your answer.

ANSWER: _____

85. A gas mixture has a total pressure of 0.985 atm. The mole fraction of O_2 is 0.209. Determine the partial pressure of the gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

86. A gas is reported to be HBr. Its density is measured as $930 \frac{\text{g}}{\text{m}^3}$ at 120 kPa and 6.63°C . Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

87. A gas cylinder is opened into a large empty flexible reservoir. The cylinder has an internal diameter of 6.36-cm, an internal height of 21-cm, and initially contains gas at 6.0 atm. The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Report your answer in SI units and justify your answer.

ANSWER: _____

88. The vapor pressure of a liquid at 61°C is 34.3 kPa. Report this pressure in SI units and justify why the numerical value changes even though the physical pressure does not.

ANSWER: _____

89. Eric is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 79.9-cm. After filling, the pressure inside the tank is 230. atm and the temperature is 305 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

90. A 654-mL sample of carbon tetrachloride (CCl_4) is initially at 472 K. Its boiling point is $77.^\circ\text{C}$ (350.15 K). The pressure is then reduced to 55% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest $^\circ\text{C}$. Justify your answer.

ANSWER: _____

91. An unknown gas has a mass of 21 g and occupies 4.4 L at 110 kPa and $20.^\circ\text{C}$. Determine the molar mass of the gas. Justify your answer. Report your answer in SI units.

ANSWER: _____

92. The reaction $2\text{NO}_2(\text{g}) \longrightarrow 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$ occurs at constant temperature and pressure. If 6.4cm^3 of NO_2 gas reacts completely, what volume of O_2 gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

93. An unknown compound is completely vaporized at 82.0°C . At a pressure of 1.2atm , the gas occupies $780.\text{mL}$ and has a mass of 0.97g . Determine the molar mass of the compound. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

94. The reaction $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 13mL of methane reacts completely, what volume of carbon dioxide gas is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

95. A sample of an ideal gas in a cylinder with a movable piston is compressed slowly while its temperature is kept constant. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

96. The reaction $\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$ occurs with all gaseous substances at the same temperature and pressure. If the reaction consumes 3.49mL of NH_3 , determine the volume of N_2 . Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

97. A compressed-gas cylinder is connected to a large evacuated storage bag. The cylinder has an internal diameter of 4.2-cm and initially contains gas at 4.47atm . After expansion to atmospheric pressure, the gas occupies a total volume of 1.90L . Determine the internal height of the cylinder. Report your answer in SI units and justify your answer.

ANSWER: _____

98. A closed-end mercury manometer connected to a gas sample has a height difference of 95 mmHg. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

99. A gas sample is trapped inside a syringe with its tip sealed. Initially the gas has a pressure of 180 kPa and occupies 7.66 L. The pressure is then increased to 5.0×10^2 kPa while the temperature remains constant. Determine the final volume. Report your answer in SI units and justify your answer.

ANSWER: _____

100. N_2 is collected over water. The total pressure of the gas sample is 730.5 torr, and the vapor pressure of water is 17 torr. Determine the partial pressure of the dry gas. Report your answer in SI units. Justify your answer.

ANSWER: _____